MIL-B-23990C
18 September 1984
SUPERSEDING
MIL-B-23990B
16 February 1979

MILITARY SPECIFICATION

BLOCKS, TACKLE, WIRE ROPE, EXTRA HEAVY DUTY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of extra heavy duty, single and multiple sheave, wire rope tackle blocks.

1.2 Classification. Wire rope tackle blocks shall be of the following class, styles, and size as specified (see 6.2.1).

Class 1 - Single sheave.
Class 2 - Double sheave.
Class 3 - Triple sheave.

Style - Rig assembly (see 3.4.3.7).

Size - The size of the block shall be specified as the outside diameter of the sheave in inches, (see table I).

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2.1), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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SPECIFICATION

MILITARY

MIL-B-3865 - Block, Rope, Tackle, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections By Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1 - Surface Texture (Surface Roughness, Waviness and Lay) (DoD Adopted).

(Applications for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A27 - Carbon-Steel Castings for General Application.
A108 - Steel Bar, Carbon, Cold Finished, Standard Quality.
A153 - Zinc Coating (Hot Dip) On Iron and Steel Hardware.
A275 - Magnetic Particle Examination of Steel Forgings.
A283 - Low and Intermediate Tensile Strength Carbon, Steel Plates, Shapes, and Bars.
A331 - Steel Bars, Alloy, Cold Finish.
A386 - Zinc Coating (Hot Dip) on Assembled Steel Products.
A611 - Steel, Cold Rolled Steel, Carbon, Structural.
A668 - Steel Forgings, Carbon and Alloy, For General Industrial Use.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The block shall be either round or oval pattern for use with wire rope and shall consist of a single sheave, double sheave, or triple sheave.

3.2 Standard commercial product. The block shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the blocks being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2.1), the contractor shall furnish one block of the same classification for first article inspection and approval (see 4.2.1 and 6.3).

3.4 Design and construction. The extra heavy duty blocks specified herein shall be designed to meet the characteristics specified in table I. The block shall be designed to prevent the rope from moving out of the sheave groove and jamming between the sheaves and the shell side or separator plate.

3.4.1 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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TABLE I. Design characteristics of extra heavy duty blocks.

Size 1/ (outside diameter of sheave in inches)	Safe working loads (tons by class 2/) 1	2	3	Wire rope diameter in inches 1/	Sheave Criteria 1/ Rim thickness in inches	Center Pin diameter in inches
6	2.50	3.00	4.00	0.375 to 0.500	1.00	0.750
8	3.50	6.00	7.00	0.500 to 0.625	1.250	0.875
10	4.00	7.00	8.00	0.625	1.250	1.00
12	5.00	10.00	12.00	0.750	1.500	1.125
14	6.50	12.00	14.00	0.750 to 0.875	1.500	1.250
16	8.00	15.00	18.00	0.875 to 1.000	1.750	1.500
18	10.00	18.00	20.00	1.000	1.750	1.500

1/ All dimensions are nominal.
2/ A ton equals 2,000 pounds.

3.4.1.1 Structural steel plate. Structural steel plate shall conform to ASTM A283, Grade D or ASTM A611, Grade C or better. The yield point of structural steel shall be not less than 33,000 pounds per square inch.

3.4.2 Processes.

3.4.2.1 Forgings. Steel forgings shall conform to ASTM A668 or better. The carbon content in steel forgings requiring welding shall be not greater than 0.35 percent. All forgings shall be of uniform quality, free of detrimental flash, scale, cracks, hard spots, cold shuts, and excessive cooling stresses. Strength and other essential physical properties of the forgings shall be adequate for the purpose intended.

3.4.2.2 Galvanizing. All metal surfaces of block and rig assembly fittings, except bearing surfaces, shall be hot dip galvanized in conformance to ASTM A386. The weight of the zinc coating shall be not less than 2 ounces per square foot.

3.4.2.3 Castings. Steel castings will conform to ASTM A27, Grade 60-30 or better to meet design requirements. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

3.4.2.4 Treatment and painting. Unless otherwise specified (see 6.2.1), the blocks shall be treated and painted in accordance with the manufacturer's

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standard practice. All surfaces of the blocks other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.4.3 Parts.

3.4.3.1 Shell side plates, separator plates, and outside straps. Shell side plates, separator plates, and outside straps shall be fabricated from structural steel plate as specified herein.

3.4.3.2 Fastening devices. Screws, pins, bolts, and similar parts subject to removal, tightening, or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed. All such parts, except bearing surfaces, shall be zinc coated in accordance with ASTM A153.

3.4.3.3 Spacers and roller guides. Spacers shall be provided, as required, to make the block a strong rugged unit with the required sheave clearances. Roller guides shall be provided, as required, to prevent the wire from coming out of the sheaves when the block is overhauled.

3.4.3.4 Sheaves. Sheaves shall be either cast steel or forged steel and shall be designed for use with the wire rope size specified in table I, and shall be designed to prevent cutting action on the rope. The rim and tread section of each sheave shall be properly grooved to fit the contour of the rope under load conditions. Overall side play shall be not greater than 0.0625 inch. Sheaves shall have a smooth-running action. Sheaves, when assembled to form blocks, shall turn freely by hand. Each sheave shall be furnished with a one-piece graphite, self-lubricating, high-grade bronze bushing, which shall be pressed into the sheave with sufficient tightness to prevent slippage under proof loads. The bushing shall be equal in length to the sheave thickness at the hub and shall have a wall thickness of not less than 0.1875 inch.

3.4.3.5 Sheave pin. The sheave pin shall be of carbon or alloy steel conforming to ASTM A108 or A331, having a carbon content of 0.25 to 0.50 percent. The center pin shall be locked in position to prevent rotation. The sheave pin shall be fitted with key head and hexagon or castellated nut with a safety pin locking device. The sheave pin shall be designed to transmit the sheave load to the shell or straps without rotating. Surface roughness of all center pin bearing surfaces shall be not greater than 63 microinches determined in accordance with ANSI B46.1.

3.4.3.6 Becket. The becket shall be formed by the extension of the strap ends, and shall be of sufficient strength to carry the loads specified.

3.4.3.6.1 Becket thimble. The block shall be furnished with a standard becket thimble. The thimble shall be commercial grade, extra heavy, wire rope type, and zinc coated.

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3.4.3.7 Rig assembly style. The block shall be furnished with a rig assembly style (see figure 1). The style shall be as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), the rig assembly shall be magnetic-particle inspected over the entire area in accordance with ASTM A275.

3.4.3.7.1 Special rig assembly styles. When specified in the contract or purchase order (see 6.2.1), special rig assembly styles not shown in figure 1 shall be provided.

3.4.4 Identification marking. Unless otherwise specified (see 6.2.1), the identification markings shall be permanently and legibly marked directly on the side of the block or on a corrosion resistant plate securely attached to the side of the block. Identification markings shall include, but not limited to, manufacturer's name or trademark, part number, wire rope diameter, and safe working load in tons.

3.4.5 Interchangeability. All blocks of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4.6 Screw threads. All screw threads shall conform to FED-STD-H28.

3.5 Performance.

3.5.1 Safe working load. The block assembly shall meet the safe working load for the size and class specified in table I. The safe working load of each block assembly, including the fittings, shall be not greater than 25 percent of the ultimate strength of the material used in its weakest component. If a commercially available block assembly is inadequate because a component (shackle or other fitting) limits its strength, the component shall be replaced with one that shall meet the strength requirements specified herein.

3.5.2 Proof load. The proof load of the block assembly shall be twice the safe working load for the size and class specified in table I and shall produce neither breakage nor permanent distortion of any component part.

3.6 Workmanship.

3.6.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.6.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.6.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.6.4 Machine work. Tolerance and gages for metal fits shall conform to the limitations specified herein and otherwise to the standards of good commercial practices.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

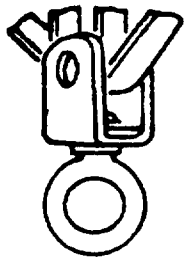
4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

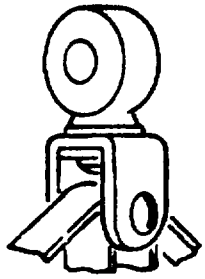
a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. The first article inspection shall be performed on one block when a first article is required (see 3.3 and 6.2.1). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified (see 6.2.1), the preproduction pack inspection of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

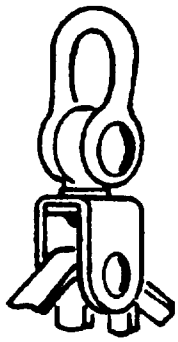
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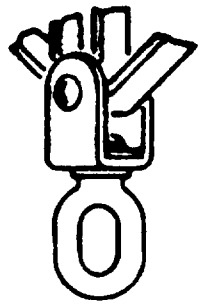
Style A
Round swivel
eye



Style B
Drilled swivel
eye



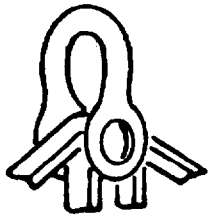
Style C
Drilled swivel
eye and shackle



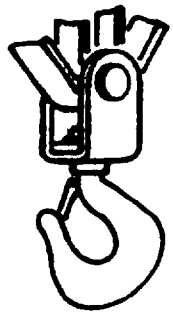
Style D
Oblong swivel
eye



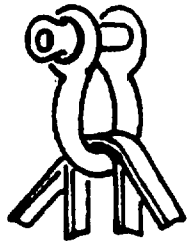
Style E
Oblong swivel
eye and shackle



Style F
Chain
shackle



Style G
Stiff swivel
safety snap hook



Style H
Upset screw pin
anchor shackle

Figure 1. Standard rig assembly styles.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All blocks offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and Acceptable Quality Level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 2.5 percent defective.

4.4 Examination. Each sample shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests.

4.5.1 Proof load test. Each sample shall be tested as follows: The block shall be attached to the fixed head of a tension testing machine or device by means of wire, rope, or round steel bar loops passed around the sheave or sheaves. The shackle or rig assembly of the block shall be attached to the moving head of the machine or the moving load by suitable links or connections. Tension shall be increased to the minimum proof load specified in 3.5.2 and shall be maintained for not less than 10 minutes. Any evidence of damage, permanent distortion, or failure while under proof load shall constitute failure of this test.

4.5.2 Magnetic-particle test. Unless otherwise specified (see 3.4.3.7 and 6.2.1), each rig assembly shall be subjected to the magnetic-particle test in accordance with ASTM A275. The rig assembly shall be rejected if the linear indication is greater than 0.125 inch in length or has a length equal to or greater than three times the width.

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4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3865. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2.1).

5. PACKING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-3865 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The extra heavy-duty blocks covered by this specification are used principally for construction and marine application. It is intended that the blocks covered by this specification should be commercial standard off-the-shelf items with available replacement parts.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class, style, and size (see 1.2 and 3.4.3.7).
c. Applicable specifications, standards, and handbooks, if other than specified (see 2.1.1).
d. When first article inspection is required (see 3.3).
e. Treatment and painting, if other than specified (see 3.4.2.4).
f. Magnetic-particle inspection, if other than specified (see 3.4.3.7 and 4.5.2).
g. Special rig assembly style, if other than shown in figure 1 (see 3.4.3.7.1).
h. Identification markings, if other than specified (see 3.4.4).
i. When a preproduction pack inspection is required and the time frame for submission (see 4.2.1 and 4.6).
j. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

(Project 3940-0176)

Review Activities:

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DLA - IS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-23990C	BLOCKS, TACKLE, WIRE ROPE, EXTRA HEAVY DUTY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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